

Asymptotic expansions



Outline

The “little o ” symbol and asymptotic expansions

Asymptotic expansions derived from the fundamental limits



The “little o ” symbol and asymptotic expansions



Introduction: the “little o ” symbol

- ▶ In this section we introduce the “**little o** ” symbol, which expresses in a compact way that a function is **negligible** with respect to another one
- ▶ We will see the link between little o and **asymptotically equivalent functions**, and how to write an **asymptotic expansion**
- ▶ We will study the calculus rules for little o (sums, products, constants) and the symbol $o(1)$
- ▶ These tools make the **computation of limits** much simpler

Definition 1: of little o

Given two functions $f(x)$ and $g(x)$, defined in a neighborhood of $c \in \mathbb{R}^$, we say that*

$$f(x) = o(g(x)) \quad \text{as } x \rightarrow c$$

(read “ $f(x)$ is little o of $g(x)$ ” as $x \rightarrow c$) if and only if

$$\frac{f(x)}{g(x)} \rightarrow 0 \quad \text{as } x \rightarrow c$$

Meaning of the little o symbol

The symbol $o(g(x))$ as x tends to c does not denote a particular function $f(x)$, but **any** function $f(x)$ such that the ratio between $f(x)$ and $g(x)$ tends to 0 as x tends to c .

Example: little o

For example, as $x \rightarrow 0$:

$$x^2 = o(x) \quad \text{since} \quad \frac{x^2}{x} \rightarrow 0$$

$$x^3 = o(x) \quad \text{since} \quad \frac{x^3}{x} \rightarrow 0$$

$$x^3 = o(x^2) \quad \text{since} \quad \frac{x^3}{x^2} \rightarrow 0$$

$$e^{-1/x^2} = o(x^4) \quad \text{since} \quad \frac{e^{-1/x^2}}{x^4} \rightarrow 0$$

Little o and asymptotically equivalent functions

- From the definition of asymptotically equivalent functions, we have:

$$f(x) \sim g(x) \text{ as } x \rightarrow c \iff \frac{f(x)}{g(x)} \rightarrow 1 \text{ as } x \rightarrow c$$

- In these cases we have, as $x \rightarrow c$:

$$\frac{f(x)}{g(x)} - 1 \rightarrow 0 \quad \text{and} \quad \frac{f(x) - g(x)}{g(x)} \rightarrow 0$$

- From the definition of little o we can then write:

$$f(x) - g(x) = o(g(x)) \text{ as } x \rightarrow c$$

Asymptotic expansion

We have the following *asymptotic expansion*:

$$f(x) \sim g(x) \text{ as } x \rightarrow c \iff f(x) = g(x) + o(g(x)) \text{ as } x \rightarrow c$$

Principal part and negligible terms

- ▶ Moreover, we have:

$$g(x) + f(x) \sim g(x) \text{ as } x \rightarrow c \iff \frac{f(x)}{g(x)} \rightarrow 0 \text{ as } x \rightarrow c,$$

since:

$$\frac{g(x) + f(x)}{g(x)} = 1 + \frac{f(x)}{g(x)}$$

which tends to 1 as $x \rightarrow c$ if and only if $f(x)/g(x)$ tends to 0 as $x \rightarrow c$.

- ▶ In such situations we say that $g(x)$ is the **principal part** of the sum $g(x) + f(x)$ and that $f(x)$ is **negligible** with respect to $g(x)$ as $x \rightarrow c$; that is:

$$f(x) = o(g(x)) \text{ as } x \rightarrow c$$

Asymptotic expansion

We have the following *asymptotic expansion*:

$$g(x) + f(x) \sim g(x) \text{ as } x \rightarrow c \iff f(x) = o(g(x)) \text{ as } x \rightarrow c$$

Little o with three functions and with products

- ▶ It follows from the definition of little o that with three functions we have:

$$f(x) = h(x) + o(g(x)) \quad \text{as } x \rightarrow c$$

$$\iff \frac{f(x) - h(x)}{g(x)} \rightarrow 0 \quad \text{as } x \rightarrow c$$

- ▶ The “little o” symbol behaves as follows with products:

$$f \cdot o(g) = o(f \cdot g)$$

$$o(f) \cdot o(g) = o(f \cdot g)$$

Example: little o and products

For example, as $x \rightarrow c$:

$$x \cdot o(x^2) = o(x^3)$$

$$\frac{o(x^3)}{x} = o(x^2)$$

$$o(x) \cdot o(x^2) = o(x^3)$$

Properties of the little o symbol

Some properties of the little o symbol, shown through examples:

$$o(x) \pm o(x) = o(x), \quad \text{as } x \rightarrow c$$

$$o(ax) = o(x), \quad \forall a \in \mathbb{R}, a \neq 0, \quad \text{as } x \rightarrow c$$

$$ao(x) = o(x), \quad \forall a \in \mathbb{R}, \quad \text{as } x \rightarrow c$$

$$o(x) + o(x^2) = o(x), \quad \text{as } x \rightarrow 0$$

$$o(x) + o(x^2) = o(x^2), \quad \text{as } x \rightarrow \infty$$

$o(1)$

Definition 2: of $o(1)$

A function f is said to be infinitesimal as $x \rightarrow c$, and we write

$$f(x) = o(1) \text{ as } x \rightarrow c \text{ if } f(x) \rightarrow 0 \text{ as } x \rightarrow c$$

Example: $o(1)$

For example:

$$\sqrt{x} = o(1) \text{ as } x \rightarrow 0^+ \quad \text{since } \frac{\sqrt{x}}{1} \rightarrow 0 \text{ as } x \rightarrow 0^+$$

$$x^{-2} = o(1) \text{ as } x \rightarrow +\infty \quad \text{since } \frac{x^{-2}}{1} \rightarrow 0 \text{ as } x \rightarrow +\infty$$

Limits and $o(1)$

- We have:

$$\lim_{x \rightarrow c} f(x) = \ell \iff \lim_{x \rightarrow c} (f(x) - \ell) = 0 \iff \lim_{x \rightarrow c} |f(x) - \ell| = 0$$

hence

$$\lim_{x \rightarrow c} f(x) = \ell \iff f(x) = \ell + o(1) \text{ as } x \rightarrow c$$

- The last expression reads “ $f(x)$ is the sum of ℓ and of a function that is infinitesimal as $x \rightarrow c$ ”

Example: $o(1)$ (1/2)

For example:

$$x^2 = x(1 + o(1)) \quad \text{as } x \rightarrow 1$$

since

$$\frac{x^2}{x} = x \rightarrow 1 \quad \text{as } x \rightarrow 1$$

$$x^2 + 3x = x(3 + o(1)) \quad \text{as } x \rightarrow 0$$

since

$$\frac{x^2 + 3x}{x} = x + 3 \rightarrow 3 \quad \text{as } x \rightarrow 0$$

Example: $o(1)$ (2/2)

$$x^2 + 3x = x^2(1 + o(1)) \quad \text{as } x \rightarrow +\infty$$

since

$$\frac{x^2 + 3x}{x^2} = 1 + \frac{3}{x} \rightarrow 1 \quad \text{as } x \rightarrow +\infty$$

$$\sqrt{x+5} = \sqrt{x}(1 + o(1)) \quad \text{as } x \rightarrow +\infty$$

since

$$\frac{\sqrt{x+5}}{\sqrt{x}} = \sqrt{1 + \frac{5}{x}} \rightarrow 1 \quad \text{as } x \rightarrow +\infty$$

Properties of the $o(1)$ symbol (1/2)

Some properties of the $o(1)$ symbol, as $x \rightarrow c$:

$$o(1) \pm o(1) = o(1)$$

$$o(1) \cdot o(1) = o(1)$$

$$c \cdot o(1) = o(1), \quad c \neq 0$$

$$(1 + o(1)) \cdot (1 + o(1)) = 1 + o(1)$$

$$\frac{1}{1 + o(1)} = 1 + o(1)$$

$$\frac{c}{1 + o(1)} - c = o(1), \quad c \neq 0$$

Properties of the $o(1)$ symbol (2/2)

The last property holds since

$$\frac{c}{1+o(1)} - c = c \left(\frac{1}{1+o(1)} - 1 \right) = c (1 + o(1) - 1) = o(1)$$



Example: $o(1)$

We have

$$(x + o(1))^2 = x^2(1 + o(1)) \quad \text{as } x \rightarrow +\infty$$

since

$$\frac{(x + o(1))^2}{x^2} \rightarrow 1 \quad \text{as } x \rightarrow +\infty$$

Summary

Little o :

- ▶ $f(x) = o(g(x))$ as $x \rightarrow c$ means $f(x)/g(x) \rightarrow 0$ as $x \rightarrow c$
- ▶ $o(g(x))$ denotes **any** function with this property

Asymptotic expansions:

- ▶ $f \sim g \iff f = g + o(g)$
- ▶ $g + f \sim g \iff f = o(g)$: g is the **principal part**, f is **negligible**

Calculus rules and $o(1)$:

- ▶ $f \cdot o(g) = o(fg)$, $o(f) \cdot o(g) = o(fg)$
- ▶ $\lim_{x \rightarrow c} f(x) = \ell \iff f(x) = \ell + o(1)$ as $x \rightarrow c$

Asymptotic expansions derived from the fundamental limits



Introduction: expansions from the fundamental limits

- ▶ In this section we translate the already known **fundamental limits** into **asymptotic expansions**
- ▶ We will obtain the expansions of $\sin x$, $\cos x$, $\log(1 + x)$, e^x and $(1 + x)^\alpha$ as $x \rightarrow 0$
- ▶ Finally we will generalize them by replacing x with any infinitesimal $\varepsilon(x)$: this is the form we will use to compute limits

Fundamental limits and asymptotic equivalences

We have seen that the following fundamental limits hold:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}, \quad \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1,$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha$$

and consequently the following **asymptotic equivalences** hold as $x \rightarrow 0$:

$$\sin x \sim x, \quad 1 - \cos x \sim \frac{1}{2}x^2, \quad e^x - 1 \sim x,$$

$$\log(1+x) \sim x, \quad (1+x)^\alpha - 1 \sim \alpha x$$

Let us now see how to obtain the **asymptotic expansions** starting from the fundamental limits.

Asymptotic expansion of $\sin x$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Hence, as $x \rightarrow 0$:

$$\frac{\sin x}{x} - 1 \rightarrow 0, \quad \frac{\sin x - x}{x} \rightarrow 0 \quad \text{hence} \quad \sin x - x = o(x)$$

which gives the asymptotic expansion:

$$\sin x = x + o(x)$$

Asymptotic expansion of $\cos x$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

Hence, as $x \rightarrow 0$:

$$\frac{1 - \cos x}{x^2} - \frac{1}{2} \rightarrow 0, \quad \frac{2(1 - \cos x) - x^2}{2x^2} \rightarrow 0$$

hence

$$2(1 - \cos x) - x^2 = o(2x^2)$$

which gives the asymptotic expansion:

$$\cos x = 1 - \frac{1}{2}x^2 + o(x^2)$$

Asymptotic expansion of $\log(1+x)$

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$$

Hence, as $x \rightarrow 0$:

$$\frac{\log(1+x)}{x} - 1 \rightarrow 0, \quad \frac{\log(1+x) - x}{x} \rightarrow 0$$

hence

$$\log(1+x) - x = o(x)$$

which gives the asymptotic expansion:

$$\log(1+x) = x + o(x)$$

Asymptotic expansion of e^x

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

Hence, as $x \rightarrow 0$:

$$\frac{e^x - 1}{x} - 1 \rightarrow 0, \quad \frac{e^x - 1 - x}{x} \rightarrow 0 \quad \text{hence} \quad e^x - 1 - x = o(x)$$

which gives the asymptotic expansion:

$$e^x = 1 + x + o(x)$$

Asymptotic expansion of $(1+x)^\alpha$

$$\lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} = \alpha \quad \text{with } \alpha \in \mathbb{R}$$

Hence, as $x \rightarrow 0$:

$$\frac{(1+x)^\alpha - 1}{x} - \alpha \rightarrow 0, \quad \frac{(1+x)^\alpha - 1 - \alpha x}{x} \rightarrow 0$$

hence

$$(1+x)^\alpha - 1 - \alpha x = o(x)$$

which gives the asymptotic expansion:

$$(1+x)^\alpha = 1 + \alpha x + o(x)$$

Generalization

- These asymptotic expansions can be generalized

If $\varepsilon(x)$ is a function that tends to zero, i.e., it is an infinitesimal (it does not matter what x tends to), we have the following **asymptotic expansions** derived from the fundamental limits.

As $\varepsilon(x) \rightarrow 0$, we have:

$$\sin(\varepsilon(x)) = \varepsilon(x) + o(\varepsilon(x))$$

$$\cos(\varepsilon(x)) = 1 - \frac{1}{2} \varepsilon^2(x) + o(\varepsilon^2(x))$$

$$\log(1 + \varepsilon(x)) = \varepsilon(x) + o(\varepsilon(x))$$

$$e^{\varepsilon(x)} = 1 + \varepsilon(x) + o(\varepsilon(x))$$

$$(1 + \varepsilon(x))^\alpha = 1 + \alpha \varepsilon(x) + o(\varepsilon(x)), \quad \alpha \in \mathbb{R}$$

Summary

From fundamental limits to asymptotic expansions (as $x \rightarrow 0$):

- ▶ $\sin x = x + o(x), \quad \cos x = 1 - \frac{1}{2}x^2 + o(x^2)$
- ▶ $\log(1+x) = x + o(x), \quad e^x = 1 + x + o(x)$
- ▶ $(1+x)^\alpha = 1 + \alpha x + o(x), \text{ with } \alpha \in \mathbb{R}$

Generalization:

- ▶ The same expansions hold when x is replaced by any infinitesimal $\varepsilon(x)$

Final summary

- ▶ The **little o** symbol expresses that a function is negligible with respect to another one: $f = o(g) \iff f/g \rightarrow 0$
- ▶ **Asymptotic equivalences** can be written as **asymptotic expansions**: $f \sim g \iff f = g + o(g)$
- ▶ The symbol $o(1)$ denotes an infinitesimal:
 $f \rightarrow \ell \iff f = \ell + o(1)$
- ▶ The **fundamental limits** give the asymptotic expansions of $\sin$, $\cos$, $\log$, $\exp$ and of powers, valid for any infinitesimal $\varepsilon(x)$